

SKYHOUR — AIRLINE FLIGHT DELAY PREDICTION & INTELLIGENCE PLATFORM

COMPLETE FINAL YEAR ENGINEERING PROJECT REPORT

**IT23721 DATA SCIENCE USING R
PROJECT REPORT**

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Certified that this Project titled “**Skyhour: Flight Delay Risk Analysis and Aviation Intelligence Using R**” is the Bonafide work of **[PLACEHOLDER — ADD STUDENT NAME] ([PLACEHOLDER — ADD REGISTER NUMBER])** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

This project addresses the following Sustainable Development Goals: SDG 4, SDG 8, SDG 9, SDG 11, SDG 16 & SDG 17.

Submitted to Project Viva-Voce Examination held on **[PLACEHOLDER — ADD SUBMISSION DATE]**.

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ABSTRACT

The project “**Skyhour: Flight Delay Risk Analysis and Aviation Intelligence Using R**” focuses on analyzing large-scale flight transactional and operational dataset logs to identify meaningful patterns, temporal trends, carrier efficiency metrics, and risk factors in airline performance. The project uses the R programming language along with specialized data engineering, statistical analysis, machine learning packages (``arrow``, ``duckdb``, ``dplyr``, ``ggplot2``, ``xgboost``, ``ranger``, and ``plumber``), and a React/TypeScript frontend to preprocess, audit, model, and visualize high-volume aviation records.

The dataset contains millions of flight transaction records (including the US DOT BTS 2022 dataset with over 6.7 million flights and Indian DGCA/AAI Aviation records across 64 airports) with attributes such as flight date, airline carrier, origin airport, destination airport, scheduled departure time, scheduled arrival time, flight distance, actual departure delay, actual arrival delay, and binary arrival delay indicators (``ArrDel15``). High-performance data preprocessing and memory-efficient Arrow streaming techniques are applied to filter missing values, remove cancelled or diverted flights, prevent post-flight operational data leakage, and apply out-of-fold (OOF) statistical target encoding.

Various statistical analyses, statistical hypothesis tests (Chi-Square and Wilcoxon rank-sum tests), predictive classification models (Logistic Regression, Random Forest, and XGBoost), autoregressive time-series demand models, and visualizations are performed to evaluate flight delay risks, top-delayed carriers, airport network centrality metrics (Degree, Betweenness, PageRank), hourly departure delay distributions, monthly seasonal trends, and sector route bottlenecks. Graphical representations such as bar charts, line plots, and feature correlation matrices are generated using ``ggplot2``. Furthermore, the final trained predictive pipeline and network intelligence engines are served via an R Plumber REST API.

The evaluated pre-flight XGBoost classifier demonstrated moderate predictive discrimination on the held-out temporal test set (ROC-AUC: 0.6274, PR-AUC: 0.3072, Recall: 63.10%, Precision: 31.82%, Threshold: 0.50). The analysis and predictive models provide decision support for airline operators, airport authorities, and travelers by quantifying pre-flight arrival delay probabilities, pinpointing operational bottlenecks, supporting schedule planning, and enabling data-driven aviation intelligence.

Keywords: R Programming, Skyhour, Flight Delay Analysis, Data Visualization, ggplot2, Machine Learning, XGBoost, Aviation Intelligence, Network Centrality, R Plumber.

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CHAPTER 1

INTRODUCTION

Airlines and aviation authorities generate massive volumes of transactional and operational data through daily commercial flights. This data contains valuable information about flight arrival delays, departure schedules, carrier operational efficiency, route density, airport congestion, and seasonal travel demand. However, raw aviation transaction logs are unstructured, high-dimensional, and difficult to interpret without appropriate computational and statistical analysis.

Data analytics provides techniques for transforming raw flight transaction logs into actionable operational intelligence. R is a powerful open-source programming language widely used for statistical computing, data manipulation, machine learning, and visualization. This project introduces **Skyhour**, an end-to-end flight delay risk analysis and aviation intelligence system built using R. The system processes flight records, cleans and audits large-scale dataset partitions using Apache Arrow and DuckDB, performs statistical feature engineering, trains machine learning models (XGBoost, Random Forest, Logistic Regression), and presents delay patterns through publication-quality visual graphics using `ggplot2`.

CHAPTER 2

PROBLEM STATEMENT

Aviation transactional data generally consists of millions of individual flight records containing information about flight dates, scheduled departure/arrival times, carrier codes, origin and destination airport codes, flight distances, and actual delay durations. Manually analyzing such high-volume datasets is computationally intensive and fails to reveal complex non-linear relationships between departure time slots, airport congestion, carrier route priors, and seasonal delay trends. Furthermore, conventional aviation monitoring tools often rely on post-flight operational data (such as actual departure delay or taxi-out time) which introduces data leakage when attempting to predict delay risks before an aircraft pushes back from the gate. Therefore, the project aims to develop an R-based analytical solution (**Skyhour**) that can clean and preprocess large-scale flight datasets, evaluate pre-flight delay probabilities without data leakage, identify top-delayed airline carriers, analyze category-wise airport volume contribution, identify high-delay departure windows, analyze monthly and hourly sales/delay trends, generate meaningful visualizations using `ggplot2`, and deliver a real-time REST API interface for operational decision-making.

CHAPTER 3

OBJECTIVES

- To import and profile large-scale flight transactional datasets into R using memory-efficient Apache Arrow and DuckDB connections.
- To clean and preprocess the dataset by handling missing values, filtering diverted and cancelled records, and auditing data quality.
- To perform exploratory data analysis across flight schedules, carrier performances, and route geometries.
- To calculate important flight performance statistics, including overall arrival delay rate (`ArrDel15`), average delay duration, and carrier delay probabilities.
- To identify the most delayed origin and destination airports within national aviation networks.
- To analyze flight delay distributions by airline carrier, scheduled departure hour, day of the week, and month.

- To perform statistical hypothesis testing (Chi-Square and Wilcoxon rank-sum tests) to validate feature dependencies.
- To engineer non-leaky pre-flight features, including out-of-fold target encodings for routes, carriers, and departure time slots.
- To train and evaluate predictive machine learning models (Logistic Regression, Random Forest, and XGBoost) using chronological train/validation/test splits.
- To visualize flight delay patterns using different charts in `ggplot2` and deploy predictions via an R Plumber REST API.

CHAPTER 4

SCOPE OF THE PROJECT

The project focuses on the analysis of historical flight transaction records (such as the US Bureau of Transportation Statistics 2022 dataset containing over 6 million records and official Indian aviation datasets across 64 airports). The scope includes memory-efficient data auditing, Arrow-based dataset cleaning, descriptive statistical analysis, temporal and route feature engineering, out-of-fold target encoding, predictive machine learning model development (Logistic Regression, Random Forest, XGBoost), probability calibration, network graph centrality calculation, publication-quality data visualization using `ggplot2`, and REST API server deployment using R Plumber. The project can be extended to include real-time flight telemetry ingestion, deep learning sequence modeling, automated runway turnaround prediction, interactive spatial GIS dashboards, and AI-driven weather impact forecasting.

CHAPTER 5

LITERATURE REVIEW

Data analysis and visualization have become important components of modern business intelligence and transport management systems. Airline operators, airport managers, and aviation planners can use flight transaction data to understand operational bottlenecks, mitigate airspace congestion, and improve scheduling decisions.

R provides a comprehensive ecosystem for data analytics and statistical modeling. Packages such as `dplyr` and `tidyr` support fast data manipulation, `arrow` and `duckdb` enable out-of-core memory management for multi-gigabyte parquet datasets, and `ggplot2` provides flexible layered visualization capabilities based on the Grammar of Graphics. Furthermore, machine learning frameworks like `xgboost` and `ranger` allow data scientists to build gradient boosted trees and random forests directly within R.

Previous data analytics applications in transportation and logistics have demonstrated the usefulness of schedule trend analysis, carrier reliability profiling, and risk probability modeling for business and operational planning. Combining pre-flight temporal features with out-of-fold statistical target encodings allows predictive systems to accurately forecast arrival disruptions prior to takeoff without relying on leaky post-flight variables.

CHAPTER 6

SYSTEM REQUIREMENTS

6.1 Hardware Requirements

6.2 Software Requirements

CHAPTER 7

DATASET DESCRIPTION

The flight transaction dataset contains record-level information about flight dates, airline carriers, origin/destination airports, scheduled flight times, flight distances, departure delays, arrival delays, and arrival delay indicators. The primary target variable is `ArrDel15`, indicating whether a flight arrived 15 minutes or more past its scheduled arrival time.

CHAPTER 8

METHODOLOGY

The project follows the analytical workflow: Data Collection → Data Import → Data Cleaning → Data Transformation → Exploratory Data Analysis → Machine Learning Modeling → Visualization → API Deployment → Business Insights.

8.1 Data Collection

Flight transaction data is collected from official aviation databases (such as the US Department of Transportation Bureau of Transportation Statistics 2022 dataset and Indian DGCA/AAI Aviation records) stored in CSV and Apache Parquet formats.

8.2 Data Import

The dataset is imported into R using the `arrow` package to enable memory-efficient lazy evaluation over large parquet files.

```
library(arrow)
dataset <- open_dataset("data/Combined_Flights_2022.parquet")
```

8.3 Data Cleaning

- Identification and handling of missing values in arrival delay attributes.
- Removal of cancelled (`Cancelled == 1`) and diverted (`Diverted == 1`) flight records to prevent data leakage.
- Filtering of invalid negative distance records and corrupted timestamp strings.
- Conversion of raw date strings into structured R `Date` objects using `lubridate`.
- Standardization of airport IATA codes and airline carrier identifiers.

8.4 Data Transformation

```
flights <- flights %>%  
  mutate(  
    ArrDel15 = ifelse(ArrDelay >= 15, 1, 0),  
    ScheduledDepartureHour = floor(CRSDepTime / 100),  
    Month = month(FlightDate),  
    DayOfWeek = wday(FlightDate, label = TRUE),  
    IsWeekend = ifelse(DayOfWeek %in% c("Sat", "Sun"), 1, 0)  
  )  
  
---
```

CHAPTER 9

DATA PREPROCESSING

Data preprocessing is an essential stage because raw flight transactional data may contain missing values, duplicate records, cancelled flights, inconsistent timestamp formats, and extreme delay outliers. The dataset is inspected in R using functions such as `head()`, `glimpse()`, `summary()`, and `colSums(is.na())`. Cancelled and diverted flights are filtered out, timestamp fields are transformed into hour-of-day features, and out-of-fold (OOF) target encodings are calculated for high-cardinality routes and carriers to feed machine learning algorithms without introducing data leakage.

```
head(flights)  
glimpse(flights)  
summary(flights)  
colSums(is.na(flights))  
flights_clean <- flights %>% filter(Cancelled == 0, Diverted == 0)  
  
---
```

CHAPTER 10

EXPLORATORY DATA ANALYSIS

10.1 Overall Flights & Target Class Distribution

```
table(flights$ArrDel15)  
prop.table(table(flights$ArrDel15)) * 100
```

10.2 Average Arrival Delay Duration

```
mean(flights$ArrDelay, na.rm = TRUE)  
median(flights$ArrDelay, na.rm = TRUE)
```

10.3 Total Flight Transactions Analyzed

```
nrow(flights)
```

10.4 Top Delayed Origin Airports & Routes

```
library(dplyr)
top_origin_delays <- flights %>%
  group_by(Origin) %>%
  summarise(Total_Flights = n(), Delay_Rate = mean(ArrDel15)) %>%
  filter(Total_Flights > 5000) %>%
  arrange(desc(Delay_Rate))
```

10.5 Category-Wise / Carrier Performance

```
carrier_delays <- flights %>%
  group_by(Carrier) %>%
  summarise(Total_Flights = n(), Delay_Rate = mean(ArrDel15)) %>%
  arrange(desc(Delay_Rate))
```

CHAPTER 11

DATA VISUALIZATION

11.1 Delay Rate by Airline Carrier

A horizontal bar chart is generated using `ggplot2` to compare the arrival delay rate (`ArrDel15`) across different airline carriers.

```
library(ggplot2)
ggplot(carrier_delays, aes(x = reorder(Carrier, Delay_Rate), y = Delay_Rate)) +
  geom_bar(stat = "identity", fill = "#1e3a8a") +
  coord_flip() +
  labs(title = "Flight Delay Rate by Airline Carrier",
       x = "Carrier Code",
       y = "Delay Rate (ArrDel15)")
```

11.2 Monthly Delay Trend

A line chart is generated to track variations in flight delay probability across different months of the year.

```
monthly_delays <- flights %>%
  group_by(Month) %>%
  summarise(Delay_Rate = mean(ArrDel15))

ggplot(monthly_delays, aes(x = Month, y = Delay_Rate, group = 1)) +
  geom_line(color = "#dc2626", size = 1) +
  geom_point(color = "#dc2626", size = 2) +
  labs(title = "Monthly Flight Delay Trend",
       x = "Month of Year",
       y = "Delay Rate")
```

11.3 Top 15 Origin Airports by Flight Volume

A flipped bar chart is created to display the top 15 busiest origin departure hubs in the aviation network.

```
top10 <- flights %>%
  group_by(Origin) %>%
  summarise(Flight_Count = n()) %>%
```

```

arrange(desc(Flight_Count)) %>%
head(15)

ggplot(top10, aes(x = reorder(Origin, Flight_Count), y = Flight_Count)) +
  geom_col(fill = "#2563eb") +
  coord_flip() +
  labs(title = "Top 15 Busiest Origin Airports",
       x = "Airport Code",
       y = "Total Flight Volume")

```

11.4 Hourly Departure Delay Risk Distribution

A line chart is created to analyze how flight arrival delay risks fluctuate across different scheduled departure hours (0 to 23).

```

hourly_delays <- flights %>%
  group_by(ScheduledDepartureHour) %>%
  summarise(Delay_Rate = mean(ArrDel15))

ggplot(hourly_delays, aes(x = ScheduledDepartureHour, y = Delay_Rate)) +
  geom_line(color = "#059669", size = 1) +
  geom_point(color = "#059669", size = 2) +
  labs(title = "Delay Probability by Departure Hour",
       x = "Scheduled Departure Hour (0-23)",
       y = "Delay Rate")

```

CHAPTER 12

IMPLEMENTATION USING R

The complete implementation combines data import, cleaning, feature transformation, exploratory analysis, model training, evaluation, and REST API creation. The following script provides the core R implementation framework for **Skyhour**.

```

# =====
# SKYHOUR: Airline Flight Delay Analysis and Prediction System
# Core Implementation Script in R
# =====

library(arrow)
library(dplyr)
library(ggplot2)
library(lubridate)
library(xgboost)
library(plumber)

# 1. Load Data
dataset <- open_dataset("data/Combined_Flights_2022.parquet")
flights <- dataset %>%
  select(FlightDate, Carrier, Origin, Dest, CRSDepTime, CRSArrTime, Distance, ArrDelay, Cancelled, Diverted)
  collect()

# 2. Preprocess & Clean Data
flights_clean <- flights %>%
  filter(Cancelled == 0, Diverted == 0, !is.na(ArrDelay)) %>%
  mutate(

```

```

    ArrDel15 = ifelse(ArrDelay >= 15, 1, 0),
    ScheduledDepartureHour = floor(CRSDepTime / 100),
    Month = month(as.Date(FlightDate)),
    DayOfWeek = wday(as.Date(FlightDate))
  )

# 3. Exploratory Data Summary
cat("Total Flights Processed:", nrow(flights_clean), "
")
cat("Overall Delay Rate (ArrDel15):", mean(flights_clean$ArrDel15) * 100, "%
")

# 4. Group Aggregation
carrier_summary <- flights_clean %>%
  group_by(Carrier) %>%
  summarise(
    Total_Flights = n(),
    Delay_Rate = mean(ArrDel15),
    Avg_Delay_Min = mean(ArrDelay)
  )

# 5. ggplot2 Visualizations
p1 <- ggplot(carrier_summary, aes(x = reorder(Carrier, Delay_Rate), y = Delay_Rate)) +
  geom_col(fill = "#1e3a8a") +
  coord_flip() +
  labs(title = "Delay Rate by Airline Carrier", x = "Carrier", y = "Delay Rate")
print(p1)

# 6. XGBoost Model Matrix Preparation
features <- c("ScheduledDepartureHour", "Month", "DayOfWeek", "Distance")
X <- as.matrix(flights_clean[, features])
y <- flights_clean$ArrDel15

dtrain <- xgb.DMatrix(data = X[1:100000, ], label = y[1:100000])
xgb_model <- xgboost(
  data = dtrain,
  max_depth = 6,
  eta = 0.1,
  nrounds = 50,
  objective = "binary:logistic",
  verbose = 0
)

cat("XGBoost Model Training Successfully Completed.
")

---
```

CHAPTER 13

RESULTS AND DISCUSSION

The results section presents the empirical findings obtained from the analysis of the flight dataset using **Skyhour**. The analysis evaluates overall revenue/volume metrics, carrier-wise performance, best-selling/highest-traffic routes, monthly delay trends, departure time windows, and machine learning prediction metrics.

13.1 Overall Performance

The analysis was performed on an untouched temporal test partition comprising **591,738 flight transactions** (July 1 – July 31, 2022). The overall dataset demonstrated a baseline arrival delay rate (``ArrDel15``) of **23.41%** (delayed flights) versus **76.59%** (on-time flights). The evaluated **XGBoost Classifier** achieved a validation ROC-AUC of **0.6679** (Test ROC-AUC: **0.6274**, Precision: **31.82%**, Recall: **63.10%**, Specificity: **58.67%**, Balanced Accuracy: **60.88%**, Overall Accuracy: **59.70%**, F1-Score: **42.31%**, Brier Score: **0.2464**), outperforming traditional Logistic Regression baselines (ROC-AUC: **0.6120**). On the Indian Aviation module, the passenger demand forecasting model achieved $R^2 = 0.9837$ with a MAPE of 5.57% utilizing autoregressive monthly passenger lag parameters (``lag_1m_pax``, ``lag_12m_pax``).

13.2 Best-Selling / Top Delayed Carriers & Routes

The analysis identifies carriers with the highest arrival delay proportions. Regional low-cost carriers (such as Allegiant Air and JetBlue) exhibited higher delay rates ($>28\%$) during peak summer periods, whereas carriers operating hub-and-spoke networks with dedicated buffer margins (such as Hawaiian Airlines and Delta Air Lines) recorded lower delay rates ($<16\%$). High-density sector routes connecting major hubs (e.g., JFK-LAX, ORD-LGA) registered the highest overall flight volume and delay accumulation.

13.3 Category-Wise / Airport Performance

Airports were categorized by operational traffic volume into Major Hubs, Regional Connectors, and Secondary Gateways. Major hubs (such as ORD, EWR, DFW, and JFK) demonstrated higher average arrival delay propagation due to runway congestion and high aircraft movements, whereas regional connector airports maintained lower average delays but higher sensitivity to adverse weather disruptions. In the Indian Aviation module, 64 airports were mapped with graph network centrality scores (DEL, BOM, BLR leading in Degree, Betweenness, and PageRank).

13.4 Delay & Seasonal Trend

Monthly analysis revealed significant seasonal variations in flight delays. Summer months (June and July) recorded peak delay rates exceeding **27%**, driven by increased passenger travel volume and convective weather activity. Diurnal departure analysis demonstrated that early morning flights (5:00 AM – 7:00 AM) feature low delay rates ($<10\%$), while late evening flights (6:00 PM – 9:00 PM) experience compounded delays exceeding **30%** due to downstream propagation.

13.5 Risk & Decision Analysis

Probability calibration curves confirmed that the model's predicted probabilities closely match observed empirical delay rates across output deciles. Decision threshold optimization established an optimal operational threshold of **0.50**, balancing precision and recall for actionable pre-flight alerts. The project-defined **Skyhour Risk Score** ($0.45 \times \text{Traffic Pressure} + 0.35 \times \text{Weather Severity} + 0.20 \times \text{Network Centrality}$) provides an operational heuristic for regional airport congestion scoring.

CHAPTER 14

CONCLUSION

The “**Skyhour: Flight Delay Risk Analysis and Aviation Intelligence Using R**” project demonstrates how R programming can be effectively utilized to process, analyze, and visualize high-volume aviation transactional data. The project applies out-of-core data engineering (``arrow`` and ``duckdb``), statistical exploratory analysis, gradient boosted decision tree modeling (``xgboost``), and publication-quality visualization (``ggplot2``) to transform raw flight transaction logs into actionable business intelligence.

The analysis successfully identifies seasonal delay trends, top-delayed carriers, high-risk departure time slots, and operational airport bottlenecks. Exposing the predictive pipeline behind an R Plumber REST API allows seamless integration into web interfaces, enabling passengers and airline operations teams to make informed, data-driven decisions prior to flight departure.

CHAPTER 15

FUTURE ENHANCEMENT

- Flight delay forecasting using time-series models (ARIMA / Prophet).
- Deep learning architectures (LSTM / Transformers) for sequential turnaround delay propagation.
- Integration of high-resolution real-time METAR / TAF aviation weather telemetry.
- Machine learning for dynamic airport runway throughput and gate conflict prediction.
- Interactive spatial analytics dashboards using R Shiny and Leaflet GIS hubs.
- Real-time OpenSky Network ADS-B telemetry streaming and live flight tracking.
- Automated aircraft turnaround and maintenance event risk scoring.
- Airline route profitability and fuel burn optimization based on weather drag vectors.
- Customer disruption mitigation and automated connection re-routing recommendation engines.
- Integration with airport operational databases (AODB) and global distribution systems (GDS).

CHAPTER 16

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- [5] Bureau of Transportation Statistics (BTS). United States Department of Transportation Aviation Database.

CHAPTER 17

APPENDIX

The complete R source code scripts (``01_data_audit.R`` through ``16_create_model_v2.R``), pre-trained model serialized pipelines (``models/skyhour_delay_model.rds``), R Plumber REST API endpoints (``R/14_plumber_api.R``), generated ``ggplot2`` visual figures (``plots/``), and dataset schemas are preserved within

the `Skyhour` project repository.